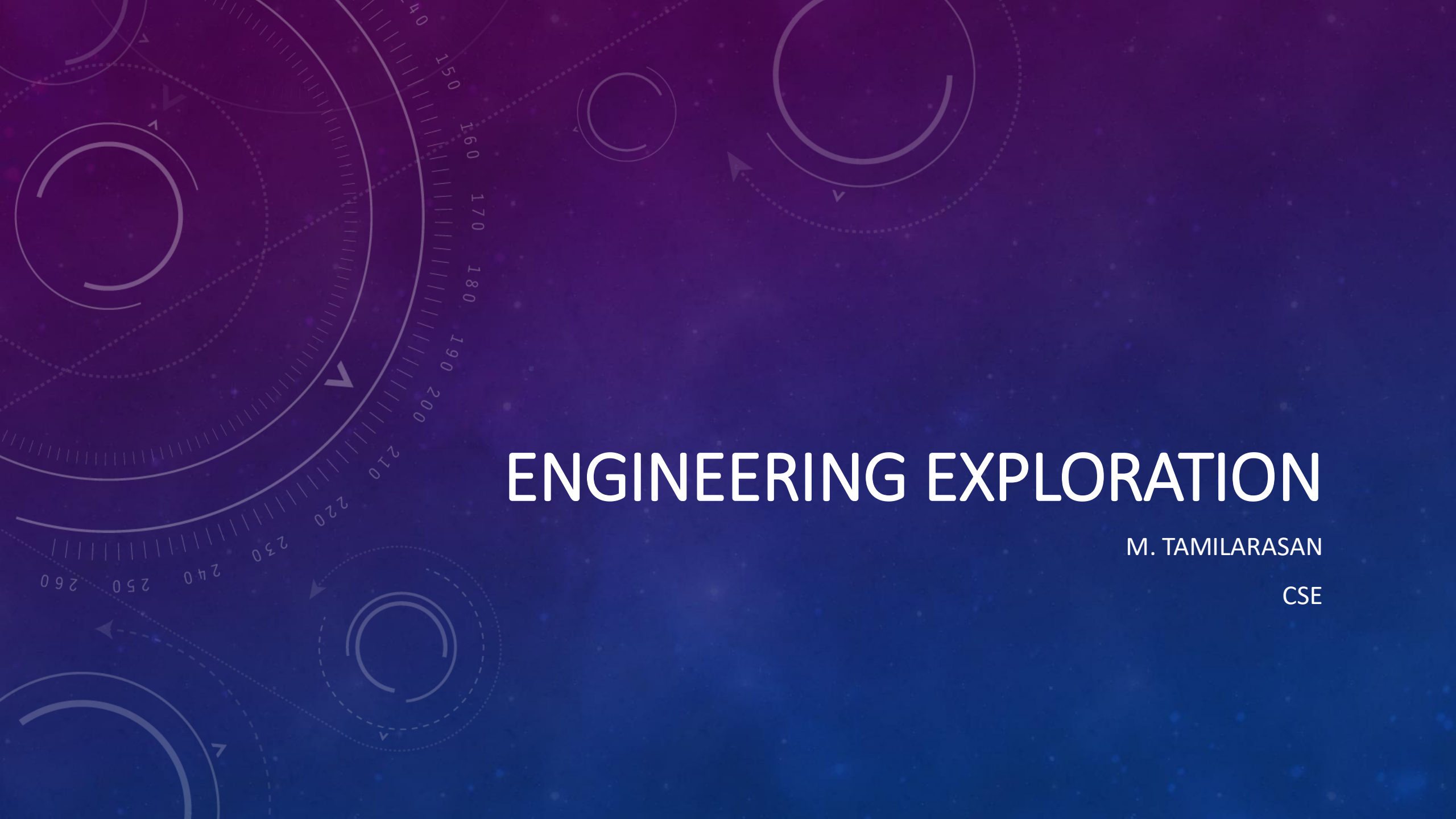


The background is a dark blue gradient with a subtle pattern of white dots. Overlaid on the left side are several technical diagrams: a large circular scale with degree markings from 140 to 260, and several smaller circular elements with arrows indicating clockwise or counter-clockwise rotation.

ENGINEERING EXPLORATION

M. TAMILARASAN

CSE

ENGINEERING

- Engineering is the application of scientific and mathematical
-
- principles to design, build,
-
- improve, and maintain machines, structures, systems, and processes that solve real- world problems.
-
- Engineers use creativity and technical knowledge to create solutions that are safe, efficient, and practical.
-
- For example, engineers design: * Bridges and buildings • Cars and airplanes * Computers and smartphones • Medical devices



SCIENCE AND ENGINEERING

- The main difference is their purpose.
-
- Science
-
- Engineering
-
- Science tries to understand how the world works.
-
- Engineering uses that knowledge to solve real-world problems.
-
- Focuses on discovering facts, laws, and theories.
-
- Focuses on designing, building, and improving products or systems



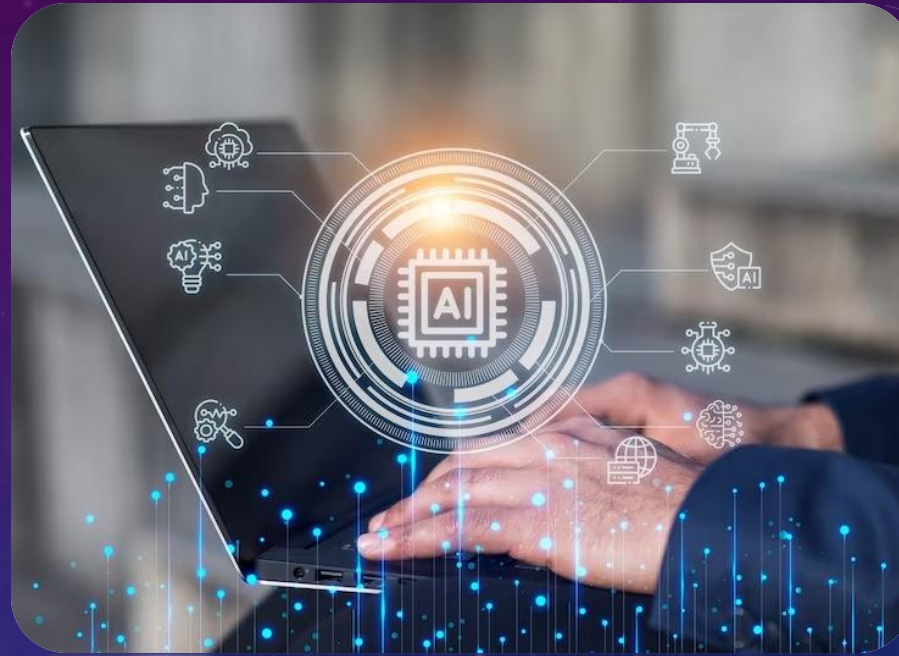
SCIENTIST AND ENGINEER

- **Scientist**
- **Engineer**
- **Studies and discovers how nature works.**
- **Uses scientific knowledge to solve practical problems.**
- **Conducts experiments and research.**
- **Designs, builds, tests, and improves products and systems.**
- **Asks “Why does this happen?” Asks “How can we make this work?” Creates new knowledge and theories. Creates useful machines, devices, software, and structures.**



NEEDS OF CSE

- Needs (Essential Requirements)
-
- These are the skills and tools every Computer Science Engineer should have:
-
- Strong programming knowledge (C, C++, Python, Java, etc.)
-
- Problem-solving and logical thinking
-
- Understanding of data structures and algorithms Knowledge of databases (SQL)



- Version control tools (Git and GitHub) Basic knowledge of operating systems, computer networks, and software engineering A laptop/computer with a good internet connection Communication and teamwork skills Continuous learning to keep up with new
-
- technologies

WANTS OF CSE

- These help a Computer Science
- Engineer grow faster and stand out: • High-paying job in a top tech company • Knowledge of Artificial Intelligence (AI) and Machine Learning (ML)
- Cybersecurity and ethical hacking knowledge
- Mobile or web development skills
- Professional certifications
- Internships and industry experience
- Work-life balance and career growth opportunities



ROLE OF CSE IN OTHER FIELDS

- Electrical Engineering
-
- Embedded systems, IoT, smart grid monitoring
-
- Smart Power Grid, Smart Energy Meter
-
- Biomedical Engineering
-
- Medical image processing, AI diagnosis, health monitoring
Wearable Health Monitor, AI-based Disease Detection

- Civil Engineering
-
- Structural analysis software, Building Information Modeling
-
- (BIM), traffic management
-
- Smart City, Smart Traffic Control System
-
- Mechanical Engineering
-
- robotics, automation, simulation
-
- Self-driving cars, Industrial robots



CASE STUDY

- The subject (e.g., Biotechnology, Engineering, Management, Nursing, etc.)
-
- The case study question (or upload a photo)
-
- Whether you want a short answer, detailed answer, or presentation format



ENGINEERING TO SOLVE WATER SUPPLY PROBLEMS IN AGRICULTURE

- PROBLEM: FARMERS IN A VILLAGE DEPENDED ON DIESEL-POWERED WATER PUMPS FOR IRRIGATION. DIESEL
- WAS EXPENSIVE, CAUSED POLLUTION, AND WAS NOT ALWAYS AVAILABLE, MAKING IRRIGATION DIFFICULT
- ENGINEERING SOLUTION: A TEAM OF ENGINEERS DESIGNED AND INSTALLED A SOLAR-POWERED WATER PUMPING SYSTEM. THE SYSTEM INCLUDED: SOLAR PANELS TO GENERATE ELECTRICITY.
- AN ELECTRIC WATER PUMP. A WATER STORAGE TANK A CONTROLLER TO MANAGE THE SYSTEM EFFICIENTLY. IMPLEMENTATION:
- SOLAR PANELS WERE INSTALLED NEAR THE FARMLAND THE PUMP DREW WATER FROM A BOREWELL ER WAS STORED IN A TANK AND DISTRIBUTED TO CROPS THROU

